

JIFENG WU

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EDUCATION

CORNELL UNIVERSITY

PhD Student

Ithaca, NY

AUGUST 2025 - EXPECTED JUNE 2030

UNIVERSITY OF BRITISH COLUMBIA

MSc in Computer Science

Vancouver, Canada

NOVEMBER 2024

WUHAN UNIVERSITY

BEng in Software Engineering, Outstanding Graduate

Wuhan, China

JUNE 2022

EXPERIENCE

AMAZON WEB SERVICES

Applied Scientist Intern, AWS Neuron

Cupertino, CA

MAY 2026 - AUGUST 2026

- Built an architecture-aware runtime rewriting framework that injects AWS Neuron Kernel Interface (NKI) kernels into vLLM under torch.compile, using structural matching and shape/correctness guards with safe fallback for incompatible configurations.
- Benchmarked NKI acceleration across 42 LLMs with complete runs on AWS Trainium2 (Trn2). Across 36 dense models, NKI achieved median 2.9× throughput while reducing time to first token and time per output token to 21% and 65% of baseline; across 6 mixture-of-experts models, it achieved median 7.8× throughput while reducing those latencies to approximately 6% and 46%.

PUBLICATIONS

Jifeng Wu, Caroline Lemieux. "QuAC: Quick Attribute-Centric Type Inference for Python." *Proceedings of the ACM on Programming Languages*, Issue OOPSLA, 2024

Jifeng Wu, Wei Li, Libing Wu, Mengting Yuan, Chun Jason Xue, Jingling Xue, Qingan Li. "Effective Stack Wear Leveling for NVM." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2023

RESEARCH

UNIVERSITY OF BRITISH COLUMBIA

QuAC: Quick Attribute-Centric Type Inference for Python

Vancouver, Canada

JANUARY 2023 - AUGUST 2024

- Designed and implemented QuAC, an attribute-centric Python type-inference system that uses attribute sets and information-retrieval techniques to predict container parameters, built-in types, and user-defined types.
- Evaluated QuAC against non-LLM and LLM baselines; outperformed non-LLM methods on container type parameters and non-built-in types while running nearly 100× faster than LLM-based methods and producing more consistent container-type predictions.

WUHAN UNIVERSITY

Effective Stack Wear Leveling for NVM

Wuhan, China

AUGUST 2021 - AUGUST 2022

- Designed Loop2Recursion, a compiler-assisted technique that transforms write-intensive loops into recursive functions to distribute stack writes and extend the lifetime of non-volatile memory.
- Implemented the transformation as an LLVM pass and built Intel Pin-based profiling and evaluation tools; demonstrated better stack wear leveling with lower performance overhead than prior approaches.

TEACHING

CORNELL UNIVERSITY

Ithaca, NY

Teaching Assistant, ECE 2400: Computer Systems Programming

JANUARY 2026 - APRIL 2026

- Developed assignment infrastructure spanning repository administration, assignment releases, automated grading workflows, final-grade scripts, and student self-checks with GitHub Actions.
- Led sections on Unix development tools, C++ environment setup, and performance profiling; mentored students in systems programming, debugging, and C/C++ development.

UNIVERSITY OF BRITISH COLUMBIA

Vancouver, Canada

Graduate Teaching Assistant, CPSC 410: Advanced Software Engineering

SEPTEMBER 2022 - DECEMBER 2022

- Mentored students on static- and dynamic-analysis projects, providing technical feedback on system design, implementation, and evaluation strategy.
- Explained advanced programming-language and software-engineering concepts through project reviews and individual guidance.

OPEN-SOURCE CONTRIBUTIONS

PYSSA

[GitHub](#)

- Developed an executable, source-oriented intermediate representation that lowers Python ASTs into explicit control flow with nested regions and basic blocks.
- Supports pluggable execution semantics for program analysis, transformation, and custom interpretation independent of Python bytecode versions.

CHATREPL

[PyPI](#)

- Built a lightweight terminal agent for OpenAI-compatible Chat Completions APIs with read, write, edit, and shell tools; automatic AGENTS.md and CLAUDE.md context discovery; and streamed tool execution.
- Supports interactive and one-shot workflows, editable multiline input, piped prompts, and image inputs.

AVL ORDER STATISTIC SET

[GitHub](#)

- Built a portable, header-only AVL set with a `std::set`-like interface, bidirectional iterators, custom comparators, and $O(\log n)$ insertion, deletion, search, rank, and selection.
- Provides a cross-platform alternative to GNU PBDS for order-statistic and range queries.

SKILLS

- Languages: Python, C, C++, Bash.
- Systems: PyTorch, vLLM, AWS Neuron/NKI, LLVM, MLIR.
- Areas: Compilers, program analysis, ML systems, formal methods.